Sample Paper 18

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

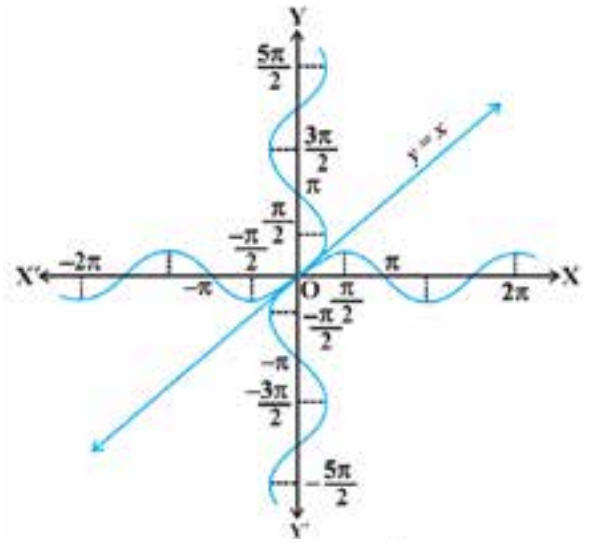
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = 1$ in the graph of $y = \text{cosec}^{-1}x$?



(a) 0

(b) $\pi/2$

(c) π

(d) Undefined

2. If $|x| \leq 1$, which of the following four is different from the other three?

(a) $\sin(\cos^{-1}x)$ (b) $\cos(\sin^{-1}x)$

(c) $\sqrt{1-x^2}$ (d) $\frac{\sqrt{1-x^2}}{x}$

3. If $f(x) = \log_e(\log_e x)$, then $f'(e)$ is equal to

(a) e^{-1}

(b) e

(c) 1

(d) 0

4. If $f(x) = \begin{cases} \frac{3\sin \pi x}{5x}, & x \neq 0 \\ 2k, & x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k is

(a) $\frac{\pi}{10}$ (b) $\frac{3\pi}{10}$

(c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{5}$

5. The area of the region bounded by the lines $y = mx$, $x = 1$, $x = 2$ and X-axis is 6 sq units, then m is equal to

(a) 3 (b) 1

(c) 2 (d) 4

6. The function $f(x) = x^2 + bx + c$, where b and c are real constants, describes

(a) One-one mapping

(b) Onto mapping

CLICK HERE TO BUY THIS BOOK FROM AMAZON

(c) Not one-one but onto mapping

(d) Neither one-one nor onto mapping

7. The equation of the curve, whose slope at any point different from origin is $y+\frac{y}{x}$, is

(a) $y = cxe^x, c \neq 0$ (b) $y = xe^x$

(c) $xy = e^x$ (d) $y + xe^x = c$

8. The differential equation representing the family of curve $y^2 = (x+\sqrt{c})$, where c is positive perimeter, is of

(a) Order 1, degree 3

(b) Order 2, degree 2

(c) Degree 3, order 1

(d) Degree 4, order 4

9. Let $\vec{a}$ and $\vec{b}$ be two non-parallel unit vectors in a plane. If the vectors $(\alpha\vec{a}+\vec{b})$ bisects the internal angle between $\vec{a}$ and $\vec{b}$, then α is equal to

(a) 1 (b) 1/2

(c) 4 (d) 2

10. If α, β, γ are the angles which a half ray makes with the positive directions of the axes, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$ is equal to

(a) 2 (b) 1

(c) 0 (d) -1

11. Which of the following triplets gives the direction cosines of a line?

(a) $<1,1,1>$ (b) $<1,-1,1>$

(c) $<1,-1,-1>$ (d) $<\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}>$

12. Solution of $\frac{dy}{dx}+y\sec x = \tan x$ is

(a) $y(\sec x+\tan x) = \sec x+\tan x - x + c$

(b) $y = \sec x+\tan x - x + c$

(c) $y(\sec x+\tan x) = \sec x+\tan x + x + c$

(d) None of the above

13. The function $f(x) = 2-3x$ is

(a) Decreasing

(b) Increasing

(c) Neither decreasing nor increasing

(d) None of the above

14. If $y = \log[\sin(x^2)]$, $0 < x < \frac{\pi}{2}$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{\pi}}{2}$ is

(a) 0 (b) 1

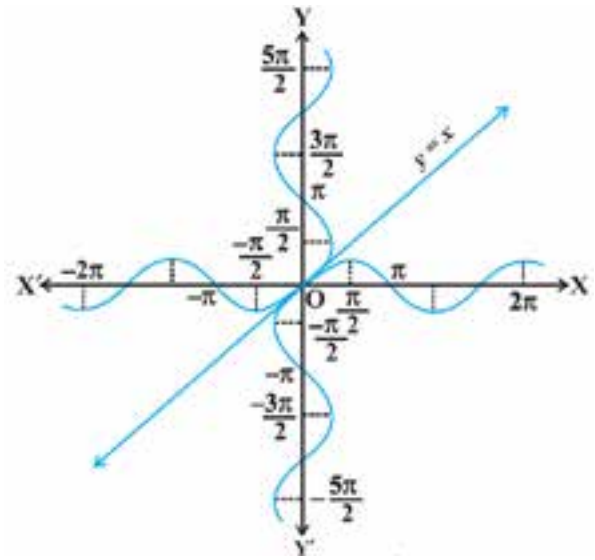
(c) $\pi/4$ (d) $\sqrt{\pi}$

15. The radius of a cylinder is increasing at the rate of 3m/s and its altitude is decreasing at the rate of 4m/s. The rate of change of volume when radius is 4m and altitude is 6m, is

(a) $80\pi m^3/s$ (b) $144\pi m^3/s$

(c) $80m^3/s$ (d) $64m^3/s$

16. Which of the following is the symmetry property of the graph $y = \tan x$ visible in the image?



(a) Symmetry about X-axis

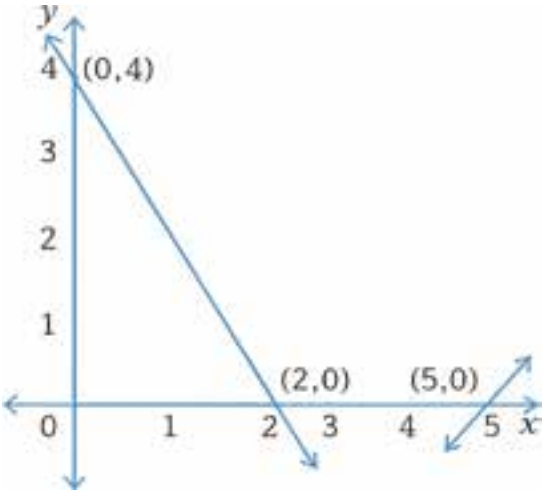
(b) Symmetry about Y-axis

(c) Symmetry about the origin

(d) No symmetry

17. Why does the minimum value of $Z = 3x+2y$ exist for this problem even though the feasible region is unbounded?

Downloaded from www.pdf.tube



(a) Because the feasible region is bounded by all constraints.

(b) Because the inequality 3x+2y<6 has no point in common with the feasible region.

(c) Because the inequality 3x+2y<6 has infinitely many points in common with the feasible region.

(d) Because every LPP always has both a minimum and maximum solution.

18. If A is a matrix of order 3 such that $A(adj\ A) = 10I$, then $|adj\ A|$ is equal to

(a) 1 (b) 10

(c) 100 (d) $10I$

19. **Assertion:** Two dice are tossed the following two events A and B are

$$A = \{(x, y): x + y = 11\},$$

$$B = \{(x, y): x \neq 5\} \text{ independent events.}$$

Reason: E_1 and E_2 are independent events, then

$$P(E_1 \cap E_2) = P(E_1)\,P(E_2)$$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. **Assertion:** If the matrix $\begin{bmatrix} 0 & a & 3 \\ 2 & b & -1 \\ c & 1 & 0 \end{bmatrix}$ is a skew symmetric matrix, then value of $(a - b - c)$ is 1

Reason: A square matrix $A = [a_{ij}]$ is said to be skew-symmetric if $A' = -A$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. Find the general solution of equation $\frac{dy}{dx} + y = 1 (y \neq 1)$

22. If $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$, then find a unit vector parallel to the vector $\vec{a} + \vec{b}$.

23. Determine the value of constant k so that the function $f(x) = \begin{cases} \frac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases}$ is continuous at $x = 0$.

24. Find the maximum value of $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cos\theta \end{vmatrix}$

25. Evaluate $P(A \cup B)$, if $2P(A) = P(B) = \frac{5}{13}$ and $P\left(\frac{A}{B}\right) = \frac{2}{5}$.

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. Find the general solution of equation $y \log y\, dx - x\, dy = 0$

27. Find $\int \frac{x-3}{(x-1)^3} e^x\, dx$.

28. Check whether the relation R in the set R of real numbers, defined by $R = \{(a, b): 1 + ab > 0\}$, is reflexive, symmetric or transitive.

29. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and I is the identity matrix of order 2, then show that $A^2 = 4A - 3I$. Hence, find A^{-1}.

CLICK HERE TO BUY THIS BOOK FROM AMAZON

30. Find the position of a point R, which divides the line joining two points P and Q whose position vectors are $2\vec{a}+\vec{b}$ and $\vec{a}-3\vec{b}$ respectively, externally in the ratio $1:2$. Also, show that P is the mid-point of line segment RQ.

31. Find the value of k, for which

$$f(x)=\begin{cases}\dfrac{\sqrt{1+kx}-\sqrt{1-kx}}{x}, & if\ -1\le x<0\\ \dfrac{2x+1}{x-1}, & if\ 0\le x<1\end{cases}$$

is continuous at $x=0$.

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. AB is the diameter of a circle and C is any point of the circle. Show that the area of ΔABC is maximum, when it is an isosceles triangle.

33. Solve graphically.
$x+y\ge 5,\ 2x+3\ge 3y,\ 0\le x\le 4,\ 0\le y\le 2$.

34. Write the following functions in the simplest from:
$\tan^{-1}\dfrac{\sqrt{1+x^2}}{x},\ x\neq 0$

35. Find the shortest distance between the lines

$$\vec{r}=\left(\hat{i}+2\hat{j}+\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+\hat{k}\right)$$

and $$\vec{r}=\left(2\hat{i}-\hat{j}-\hat{k}\right)+\mu\left(2\hat{i}+\hat{j}+2\hat{k}\right)$$

SECTION - E

Section E consists of 3 case study based questions of 4 marks each.

36. Goods and Services Tax (GST) is an indirect tax (or consumption tax) used in India on the supply of goods and services. It is a comprehensive, multistage, destination-based tax: comprehensive because it has subsumed almost all the indirect taxes except a few state taxes. Multi-staged as it is, the GST is imposed at every step in the production process, but is meant to be refunded to all parties in the various stages of production other than the final consumer and as a destination-based tax, it is collected from point of consumption and not point of origin like previous taxes.



A GST form is either filed on time or late, is either from a small or a large business, and is either accurate or inaccurate. There is an 11% probability that a form is from a small business and is accurate and on time. There is a 13% probability that a form is from a small business and is accurate but is late. There is a 15% probability that a form is from a small business and is on time. There is a 21% probability that a form is from a small business and is inaccurate and is late.

(i) If a form is from a small business and is accurate, what is the probability that it was filed on time?

(ii) What is the probability that a form is from a large business?

37. Fertilizer, natural or artificial substance containing the chemical elements that improve growth and productiveness of plants. Fertilizers enhance the natural fertility of the soil or replace chemical elements taken from the soil by previous crops.

Downloaded from www.pdf.tube



The following matrix gives the proportionate mix of constituents used for three fertilisers:

$$\begin{array}{r} \\ \text{I} \\ \text{Fertilisers II} \\ \text{III} \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0.5 & 0 & 0.5 & 0 \\ 0.2 & 0.3 & 0 & 0.5 \\ 0.2 & 0.2 & 0.1 & 0.5 \end{bmatrix} \end{array} \quad \text{Constituents}$$

(i) If sales are 1000 tins (of one kilogram) per week, 20% being fertiliser I, 30% being fertiliser II and 50% being fertiliser III, how much of each constituent is used.

(ii) If the cost of each constituents is Rs 5, Rs 6, Rs 7.5 and Rs 10 per 100 grams, respectively, how much does a one kilogram tin of each fertiliser cost

(iii) What is the total cost per week?

38. Bata India is the largest retailer and leading manufacturer of footwear in India and is a part of the Bata Shoe Organization. Incorporated as Bata Shoe Company Private Limited in 1931, the company was set up initially as a small operation in Konnagar (near Calcutta) in 1932. In January 1934,



The manager of BATA show room at Jaipur determines that the price p (dollars) for each pair of a popular brand of sports sneakers is changing at the rate of

$$p'(x) = \frac{-300x}{(x^2+9)^{3/2}}$$

when x (hundred) pairs are demanded by consumers. When the price is \$75 per pair, 400 pairs $(x = 4)$ are demanded by consumers.

(i) Find the demand (price) function $p(x)$.

(ii) At what price will 500 pairs of sneakers be demanded? At what price will no sneakers be demanded?

(iii) How many pairs will be demanded at a price of \$90 per pair?

CLICK HERE TO BUY THIS BOOK FROM AMAZON